SAPTHAGIRI NPS UNIVERSITY
UNMATCHED EXCELLENCE, UNLIMITED POTENTIAL

Department of Physics
Question Bank
Semester II Academic Year 2025/26
Module 1 & 2

Module 1 Elements of Quantum Mechanics

1. What is a matter wave? Define De Broglie wavelength. Derive an expression for de Broglie wavelength in terms of momentum.
2. Derive an expression for de Broglie wavelength in terms of kinetic energy.
3. State de Broglie's Hypothesis. Show that the de Broglie wavelength of an electron accelerated by a potential is found to be equal to $\frac{12.28}{\sqrt{V}}$ Å.
4. Derive an expression for de Broglie wavelength in terms of kinetic energy and accelerating potential.
5. State and explain Heisenberg's uncertainty principle. Give its significance.
6. What is a wave function? Discuss the properties of wave function with graph.
7. Develop Schrodinger's time independent time equation for one dimension.
8. What is the de Broglie wavelength of a proton whose energy is 3eV? Given the mass of proton =1.67 X 10^{-27} Kg.
9. Calculate the de Broglie wavelength associated with an electron accelerated through a potential difference of 2kV.
10. Calculate the kinetic energy of an electron of wavelength 18nm.
11. Calculate the de Broglie wavelength of an electron moving with 1/10th part of the velocity of the light.
12. An electron has a speed of 4.8 x 105 m/s accurate to 0.012%. With what accuracy can the position of electron be located?
13. An electron has a speed of 100m/s. The inherent uncertainty in its measurement is 0.005%. What will be the uncertainty that arises in the measurement of its position?
14. In a measurement of position and velocity of an electron moving with the speed of
15. 6x105m/s, calculate the highest accuracy with which its position could be determined if the inherent error in the measurement of its velocity is 0.01% for the speed stated.

16. Calculate the energy in eV for the first excited state of an electron in an infinite potential well of width 2Å.

17. Calculate the zero-point energy for an electron in a box of width 10Å.

18. An electron is bound in 1-D potential well of width 0.18nm, find its energy value in eV in the second excited state.

19. The ground state energy of an electron in an infinite well is 5.6 X 10-3 eV. What will be the ground state energy if the width of the well is doubled?

20. Calculate the de Broglie wavelength of a neutron whose energy is 5 eV.

21. Calculate the kinetic energy of an electron with a wavelength of 15 nm.

22. An electron has a speed of 3.2 × 105 m/s accurate to 0.01%. What is the accuracy with which its position can be measured?

23. Calculate the energy in eV for the first excited state of an electron in an infinite potential well of width 1.5 Å.

24. Find the ground state energy of an electron in a box of width 8 Å.

25. Calculate the energy in eV for the second excited state of an electron in an infinite potential well of width 30 Å.

Module 2: Introduction to Quantum Computing

1. What are qubits? Write 5 properties of qubits.

2. State and explain Moore's law with graph and discuss its end.

3. What limitations are currently being faced in continuing Moore's Law? What is meant by the term "end of Moore's Law"?

4. What is qubit? Explain its representation in detail.

5. Explain Quantum Superposition and Quantum entanglement.

6. What is conjugate of a matrix, transpose of a matrix, unitary matrix? Write the matrix form of each matrix.

7. Give the Definition, matrix form, operations and truth table for .Pauli's X gate, Y gate and Z gate.

8. Give the matrix form, operations and truth table for Hadamard gate.

9. Give the matrix form, operations and truth table for S gate and T gate.
10. What is controlled NOT gate (CNOT gate)? Give its matrix form and truth table. Discuss the 4 operations of CNOT gate.
11. What is swap gate? Write its matrix form, symbol and truth table.
12. What is controlled z gate? Write its matrix form, symbol and truth table.
13. What is Toffoli gate? Write its matrix form, symbol and truth table.
14. Find the transpose of the given matrix. $\mathbf{A} = \begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{4} & \mathbf{5} & \mathbf{6} \\ \mathbf{7} & \mathbf{8} & \mathbf{9} \end{bmatrix}$
15. Given $A = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ prove that $A = A^{\dagger}$
16. Show that the Matrix $U = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix}$ is Unitary.
17. Verify the given matrix is unitary or not $\mathbf{A} = \begin{bmatrix} \mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{i} \end{bmatrix}$
18. Find the complex conjugate $\mathbf{A} = \begin{bmatrix} \mathbf{1} & \mathbf{x+iy} \\ \mathbf{x-iy} & \mathbf{i} \end{bmatrix}$
19. Verify the given matrix is unitary or not $\mathbf{U} = \frac{\mathbf{1}}{\mathbf{2}}\begin{bmatrix} \mathbf{1+i} & \mathbf{1-i} \\ \mathbf{1-i} & \mathbf{1+i} \end{bmatrix}$
20. Show that the Pauli X gate acts as classical NOT gate.
21. Show how the Hadamard gate transforms $|\mathbf{0}\rangle$ and $|\mathbf{1}\rangle$
22. Explain the output with a step-by-step derivation when the Hadamard gate is applied twice on $|\mathbf{0}\rangle$
23. Show that Hadamard Gate is Unitary.
24. Show that S gate can be formed by connecting two T gates in Series.
25. Show that the Pauli Y gate can be constructed using a combination of X and Z gates.